Reg. No. | | | | | | | | | | | [18MTCSE201]

SECOND SEMESTER M.TECH EXAMINATION, JUNE 2019

OPERATING SYSTEM AND DESIGN PRINCIPLES

Time: 03 Hours **Maximum Marks:100**

Instructions: **i. Answer FIVE full questions. Questions 1 and 2 are compulsory.**
ii. Draw neat sketches wherever necessary.

1.	**a.**	Mention the operating system objectives and functions.	**10**
	b.	Explain in detail the readers/writers problem.	**10**
2.	**a.**	What is deadlock? Explain about deadlock detection and avoidance.	**10**
	b.	Explain with diagram of various memory partitioning techniques.	**10**
3.	**a.**	Describe the different types of processor scheduling. What criteria affect the scheduler performance?	**10**
	b.	Mention the advantages of DMA? Explain various DMA configurations.	**10**
		OR	
4	**a.**	Discuss about the RAID structure in disk management with various RAID levels of organization in detail.	**10**
	b.	Illustrate the importance of file system security.	**10**
5.	**a.**	What is an embedded system? Give its organization and unique characteristics.	**10**
	b.	List the goals of Tiny Os? Explain how it helps WSNs with an example application and configuration.	**10**
		OR	
6.	**a.**	Give the virus classification. Explain how they infect files and their countermeasures.	**10**
	b.	What is a rootkit? Explain with diagram how it impacts system performance.	**10**
7.	**a.**	Discuss in detail about DAC and RBAC.	**10**
	b.	What is a digital immune system? Discuss.	**10**
		OR	
8.	**a.**	Mention the need for middleware? Discuss the role of middleware in client/server architecture.	**10**
	b.	List the benefits and disadvantages of blocking and nonblocking primitives for message passing.	**10**

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